Closeness is equivalent to the distance inequality on the Cauchy reals

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'RC-sim-eqv-le' (source lines 1578-1582, Theorem 11.3.44); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's openness lemma for the order, whose first item is invoked twice inside the retained proof; it is the immediately preceding numbered result of the same section, so the retained sequence keeps both numbers.

Editorial note. The retained counter values (chapter 11, section 3, theorem offset 42) make the retained context print as Theorem 11.3.43 and the counted theorem as Theorem 11.3.44, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. No retained line refers to a label, citation or figure outside the retained spans, so the package declares no external reference.

Editorial scope. One original result of the HoTT Book, counted once: closeness is the distance inequality on the Cauchy reals (Theorem 11.3.44).

Allowed standard prerequisites (editorial).

- Lipschitz properties of subtraction and absolute value. Subtraction and absolute value on the Cauchy reals are Lipschitz for closeness, so closeness is preserved by them.. They are used at source lines 1584 and 1599.
- Roundedness and the triangle inequality for closeness. closeness is rounded and satisfies the triangle inequality.. They are used at source lines 1589-1590 and 1602.
- Order density of the rationals in the Cauchy reals. Between any two related rationals there is a rational strictly between them.. It produces theta < epsilon at source line 1598.

Lemma 11.3.43. *Suppose* $q : \mathbb{Q}$ *and* $u : \mathbb{R}_{\mathsf{c}}$ *satisfy* $u < \mathsf{rat}(q)$. *Then:*

(i) For any $v : \mathbb{R}_{\mathsf{c}}$ *and* $\epsilon : \mathbb{Q}_+$, *if* $u \sim_\epsilon v$ *then* $v < \mathsf{rat}(q + \epsilon)$.

(ii) There exists $\epsilon : \mathbb{Q}_+$ *such that for any* $v : \mathbb{R}_{\mathsf{c}}$, *if* $u \sim_\epsilon v$ *we have* $v < \mathsf{rat}(q)$.

Theorem 11.3.44. $(u \sim_\epsilon v) \simeq (|u - v| < \mathsf{rat}(\epsilon))$ *for all* $u, v : \mathbb{R}_{\mathsf{c}}$ *and* $\epsilon : \mathbb{Q}_+$.

Proof. The Lipschitz properties of subtraction and absolute value imply that if $u \sim_\epsilon v$, then $|u - v| \sim_\epsilon |u - u| = 0$. Thus, for the left-to-right direction, it will suffice to show that if $u \sim_\epsilon 0$, then $|u| < \mathsf{rat}(\epsilon)$. We proceed by $\mathbb{R}_\mathsf{c}$-induction on u.

If u is rational, the statement follows immediately since absolute value and order extend the standard ones on $\mathbb{Q}_+$. If u is $\mathsf{lim}(x)$, then by roundedness we have $\theta : \mathbb{Q}_+$ with $\mathsf{lim}(x) \sim_{\epsilon-\theta} 0$. By the triangle inequality, therefore, we have $x_{\theta/3} \sim_{\epsilon-2\theta/3} 0$, so the inductive hypothesis yields $|x_{\theta/3}| < \mathsf{rat}(\epsilon - 2\theta/3)$. But $x_{\theta/3} \sim_{2\theta/3} \mathsf{lim}(x)$, hence $|x_{\theta/3}| \sim_{2\theta/3} |\mathsf{lim}(x)|$ by the Lipschitz property, so Lemma 11.3.43(i) implies $|\mathsf{lim}(x)| < \mathsf{rat}(\epsilon)$.

In the other direction, we use $\mathbb{R}_\mathsf{c}$-induction on u and v. If both are rational, this is the first constructor of $\sim$.

If u is $\mathsf{rat}(q)$ and v is $\mathsf{lim}(y)$, we assume inductively that for any ϵ, δ, if $|\mathsf{rat}(q) - y_\delta| < \mathsf{rat}(\epsilon)$ then $\mathsf{rat}(q) \sim_\epsilon y_\delta$. Fix an ϵ such that $|\mathsf{rat}(q) - \mathsf{lim}(y)| < \mathsf{rat}(\epsilon)$. Since $\mathbb{Q}$ is order-dense in $\mathbb{R}_\mathsf{c}$, there exists $\theta < \epsilon$ with $|\mathsf{rat}(q) - \mathsf{lim}(y)| < \mathsf{rat}(\theta)$. Now for any δ, η we have $\mathsf{lim}(y) \sim_{2\delta} y_\delta$, hence by the Lipschitz property

$$|\mathsf{rat}(q) - \mathsf{lim}(y)| \sim_{\delta+\eta} |\mathsf{rat}(q) - y_\delta|.$$

Thus, by Lemma 11.3.43(i), we have $|\mathsf{rat}(q) - y_\delta| < \mathsf{rat}(\theta + 2\delta)$. So by the inductive hypothesis, $\mathsf{rat}(q) \sim_{\theta+2\delta} y_\delta$, and thus $\mathsf{rat}(q) \sim_{\theta+4\delta} \mathsf{lim}(y)$ by the triangle inequality. Thus, it suffices to choose $\delta :\equiv (\epsilon - \theta)/4$.

The remaining two cases are entirely analogous. □